

Angel Tu

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EDUCATION

University of California, Berkeley

Berkeley, CA

Bachelor's degree in Computer Science, GPA: 3.684 / 4.0

Aug 2021 - May 2025

Courses: Efficient Algorithms, Data Structures, Operating Systems, Database Systems, Computer Architecture, Distributed Systems, Internet Architecture and Protocols, Computer Security, Artificial Intelligence, Computer Simulations, Computer Graphics, Quantum Computing, Full Stack Web Development, OOD, System Design.

- Physics and Astronomy Scholars Program (PA Scholars).
- Academic Internship for EECS Department - Assisting teaching at course CS 61B.

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++, C#, JavaScript, HTML & CSS, Swift, Go, SQL, Scheme, RISC-V, MATLAB, R, Ruby.

Frameworks: SwiftUI, React, Node.js, Spring Boot, Google Cloud Platform, RESTful API, Linux, iOS, LangChain, NumPy, SciPy, Matplotlib, AWS.

Databases and Tools: MySQL, MongoDB, SQLite, Firebase, Xcode, NoSQL, Pandas, Git, Maven, Docker.

WORK EXPERIENCE

Caraio Technologies

Sacramento, CA

Software Engineer Intern

August 2024 - January 2025

- Designed and developed a cutting-edge platform in **Python** to assist in diagnosing chromosomal disorders.
- Leveraged AI-driven techniques to analyze and interpret cytogenetic data.

Housemint

San Jose, CA

Software Engineer Intern

May 2024 - Sep 2024

- Designed and developed user-friendly front-end interface using **Swift**.
- Designed and built 2 **RESTful API** endpoints using **Java**, **Spring Boot**, **Maven**, and **GCP**.
- Optimized database schema design of 3 tables storing user data and media files using **MySQL**.
- Analyzed data getting from using **Apple ARKit**, and created 3D room scans using Swift.
- Created 2D floor plan using 3D USDZ models from using **Apple RoomPlan API**.

Ai.vocate

Sacramento, CA

Machine Learning Engineer Intern

May 2023 - Aug 2023

- Developed a chatbot that answers customers' legal related questions for the company's iOS app in Python.
- Built a **RAG** based ChatBot in **Python** by embedding legal text documents to vectors and using **Meta's FAISS similarity search** to retrieve the relevant embeddings to answer customer's queries.

PROJECTS

Paws Up

May 2022 - Jan 2023

Github Link: <https://github.com/theo-ranger/Paws-up>

- Built an instagram-like **iOS mobile app** using **Swift** and **SwiftUI** where the users can create posts to share their love of animals and make special posts if they find a lost pet or have lost a pet themselves.
- Implemented the **MVVM** architecture, and used **Firebase** to store the users' information and media files.

Computer Graphics and Imaging

Jan 2024 - May 2024

Demo Link: https://cal-cs184-student.github.io/hw-webpages-sp24-lycorisradiatu/final_project/index.html

- Implemented various **subdivision** schemes and **remeshing** using **C++**.
- Implemented the modified butterfly subdivision scheme and the **sqrt3**-subdivision.
- Implemented isotropic remeshing algorithm based on our vertex shift and edge collapse algorithm.

Game Design and Development

Jan 2022 - May 2022

Demo Link: <https://circlecly.itch.io/spermulator>

- Developed a video game to simulate the fertilization process in biology using the **Unity Engine** and **C#**.

RESEARCH

Undergraduate Laboratory at Berkeley - Physics and Astronomy Lab

- Used Doppler Spectroscopy to Find Binary Systems with Protoplanetary Disks.
- Found binary systems using a dataset that contains stellar spectra of 28 young stars gathered using the APF.
- Used **pandas**, **xarray**, and **scipy.stats** to clean the data and generate the simulations of star movements.